

From Prompts to Review Harnesses

Scaling AI Code Reviews
Across 70+ Software Projects

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CGI





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Agenda

1 Why?

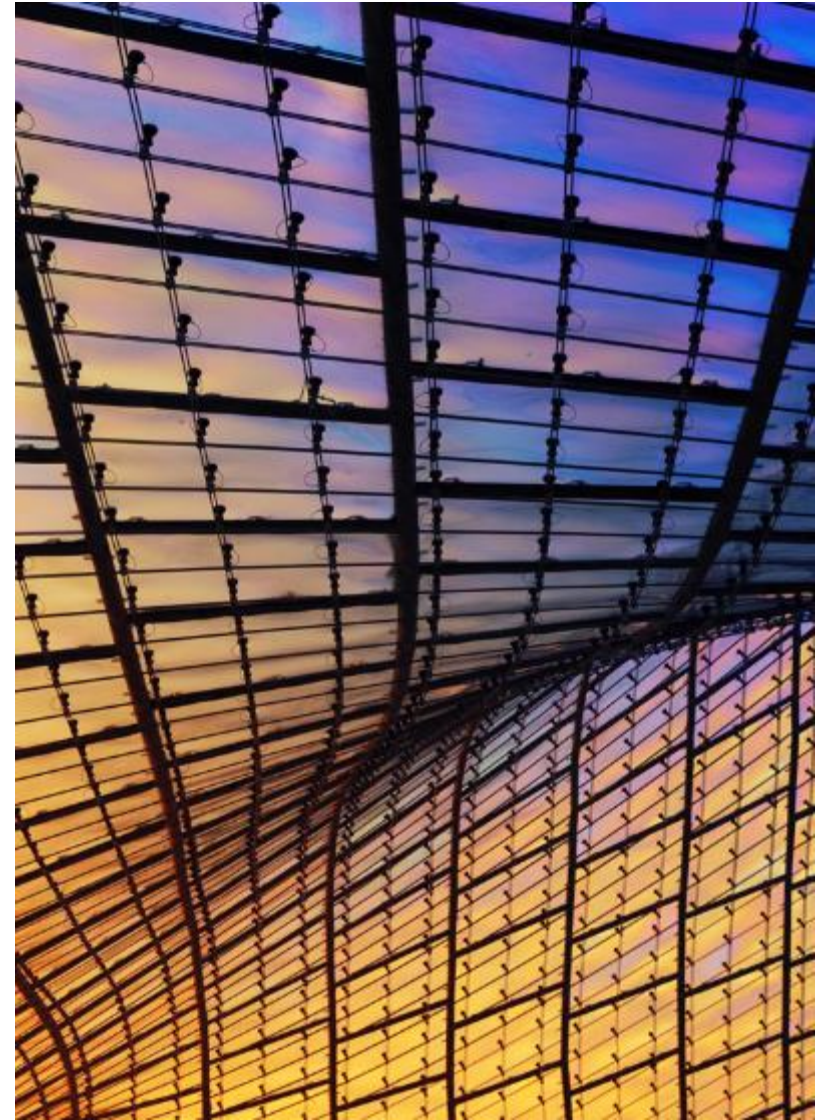
2 Initial Approach

3 Problems & Solutions

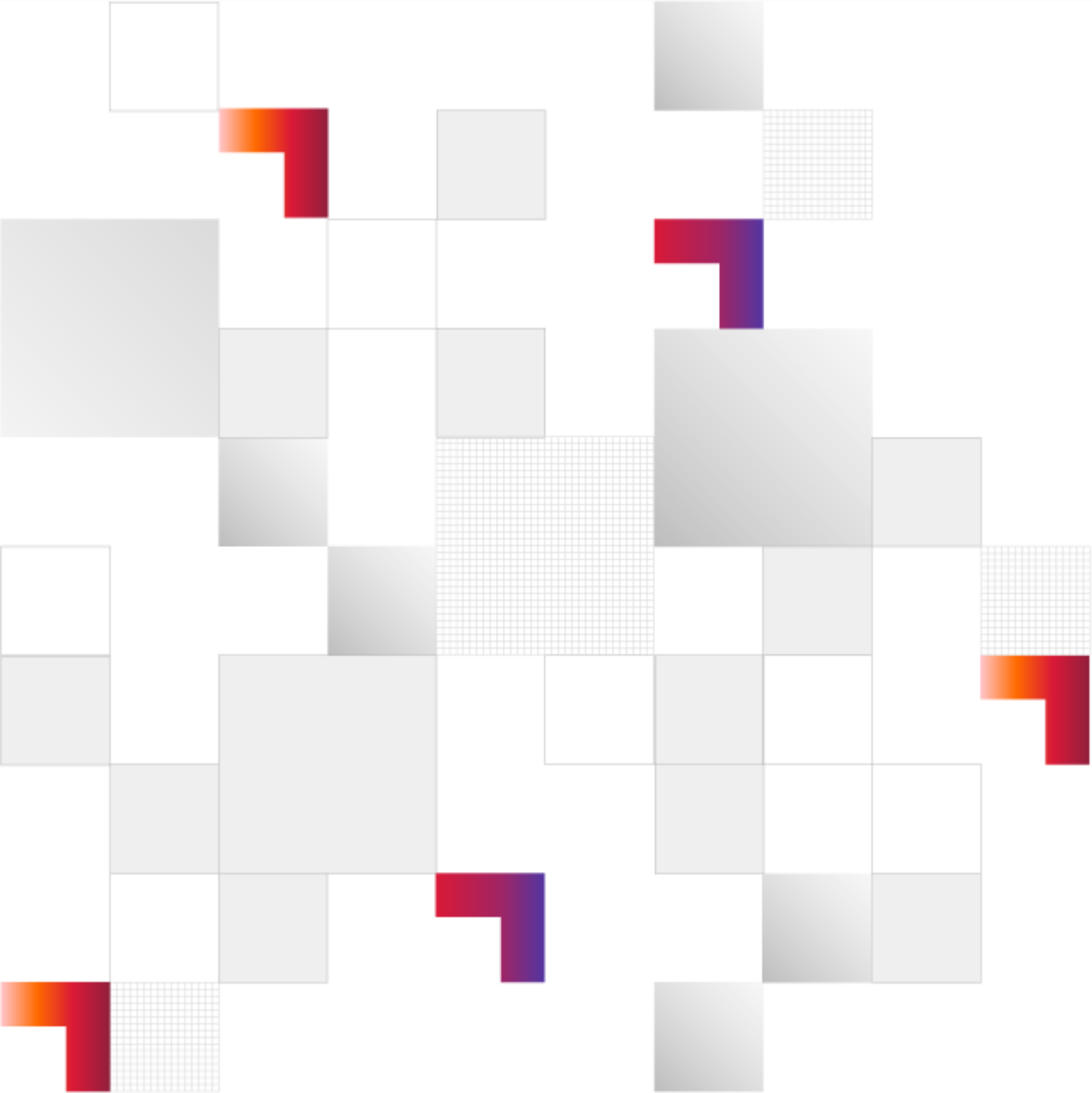
4 Results

5 Learnings

6 Next Steps



Why?



Why AI Reviews?

- critical manual findings in one project
- as a result, we needed to review 70+ more projects
- not enough budget or people for those manual reviews

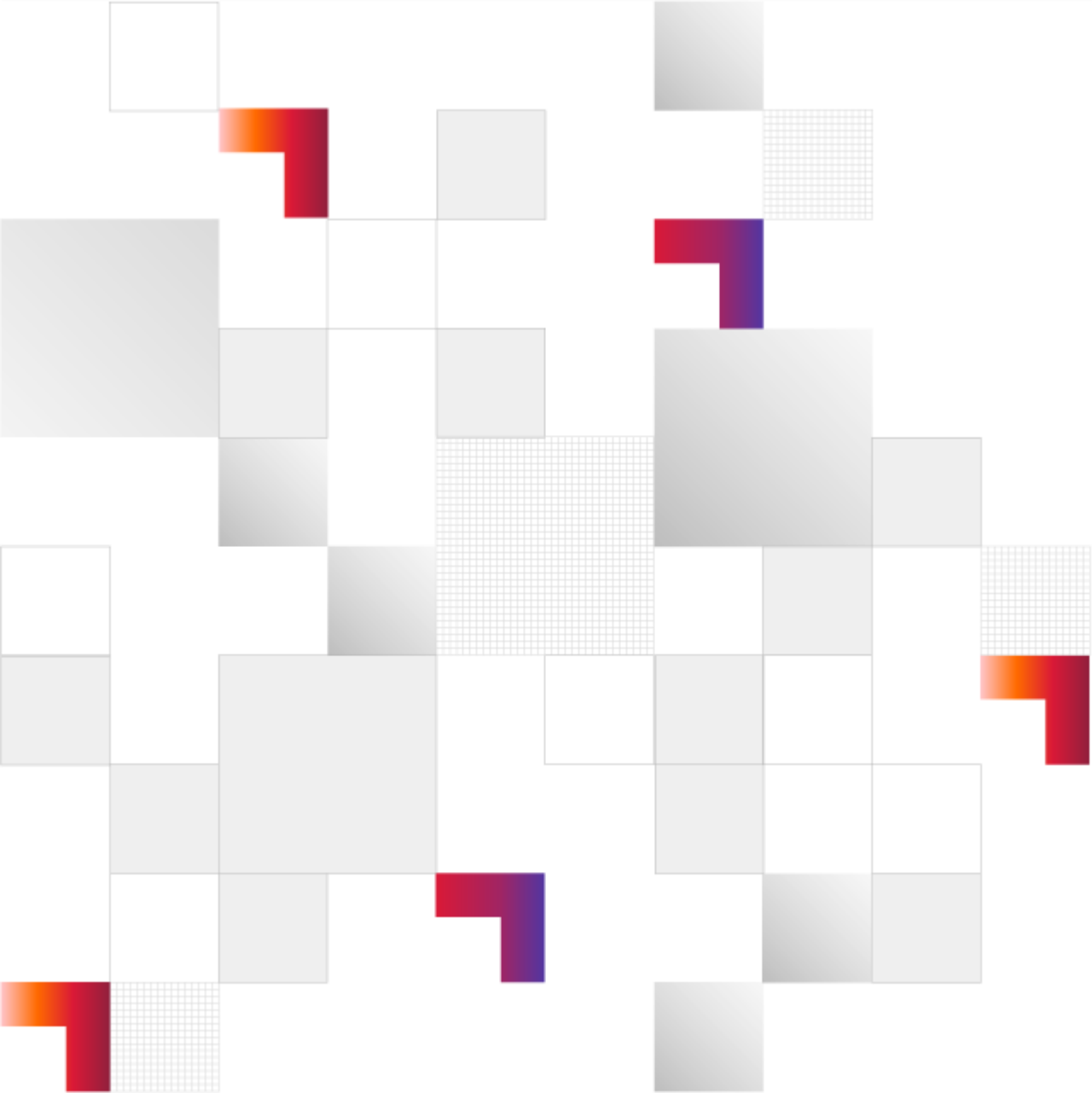


Why Build Our Own Tool?

- Compliance
- better long-term cost control
- complete model independence
- better control over review criteria



Initial Approach

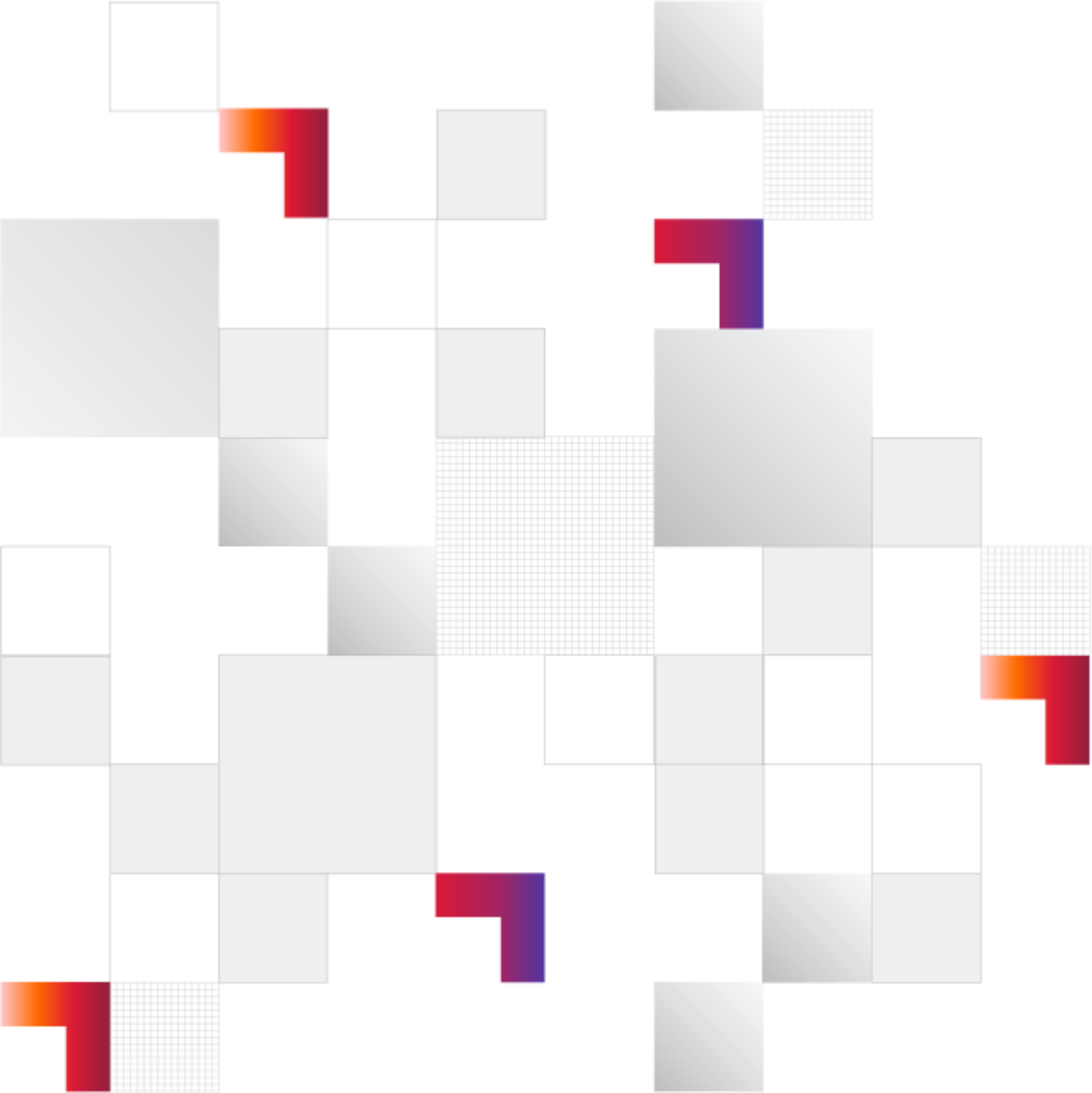


Prompt-Based Approach

- OpenCode as agent runner
 - one prompt
 - one detailed task Markdown
 - multiple checklists
 - one report template
- result looked good at first



Problems & Solutions



Prompt-Based Approach: Problems

- agents just wouldn't perform consistent, complete reviews
 - sub-agents helped, but did not solve the problem
 - highlight: review of a 700-file project by reading “11 key files” and extrapolating the results
- report template was never fully followed



Prompt-Based Approach: More Problems

- Claude Sonnet 4.5 → 4.6 suddenly started actively ignoring large parts of the repo
- trying to force it through the prompt is simply not reliable



Prompt-Based Approach: AI Rationalization



“I understood that the task was to look at everything - but decided that the review was good enough after looking at half of it.”

“I consider looking at these files irrelevant in the context of the review, even though that clearly violates the rules.”

Determinism and Enforcement: Core Idea

- deterministic review process mapped out with Strands Agents
- implement programmatically solvable steps programmatically
- each AI step only has the tools necessary for that step
- without successful validation, the process does not move to the next step



Determinism and Enforcement: Solving the Problems

- programmatic control over checklists and files reduces hallucinations and “forgetting” tasks
- completeness is enforced by requiring the individual results to be handed over to the programmatic process
- report generated programmatically based on the results; generation is rejected if results are missing



Determinism and Enforcement: Behind the Scenes

- carefully designed storage of results in memory files
 - raw data behind the report
 - goals: resuming interrupted reviews, incremental reviews, later processing

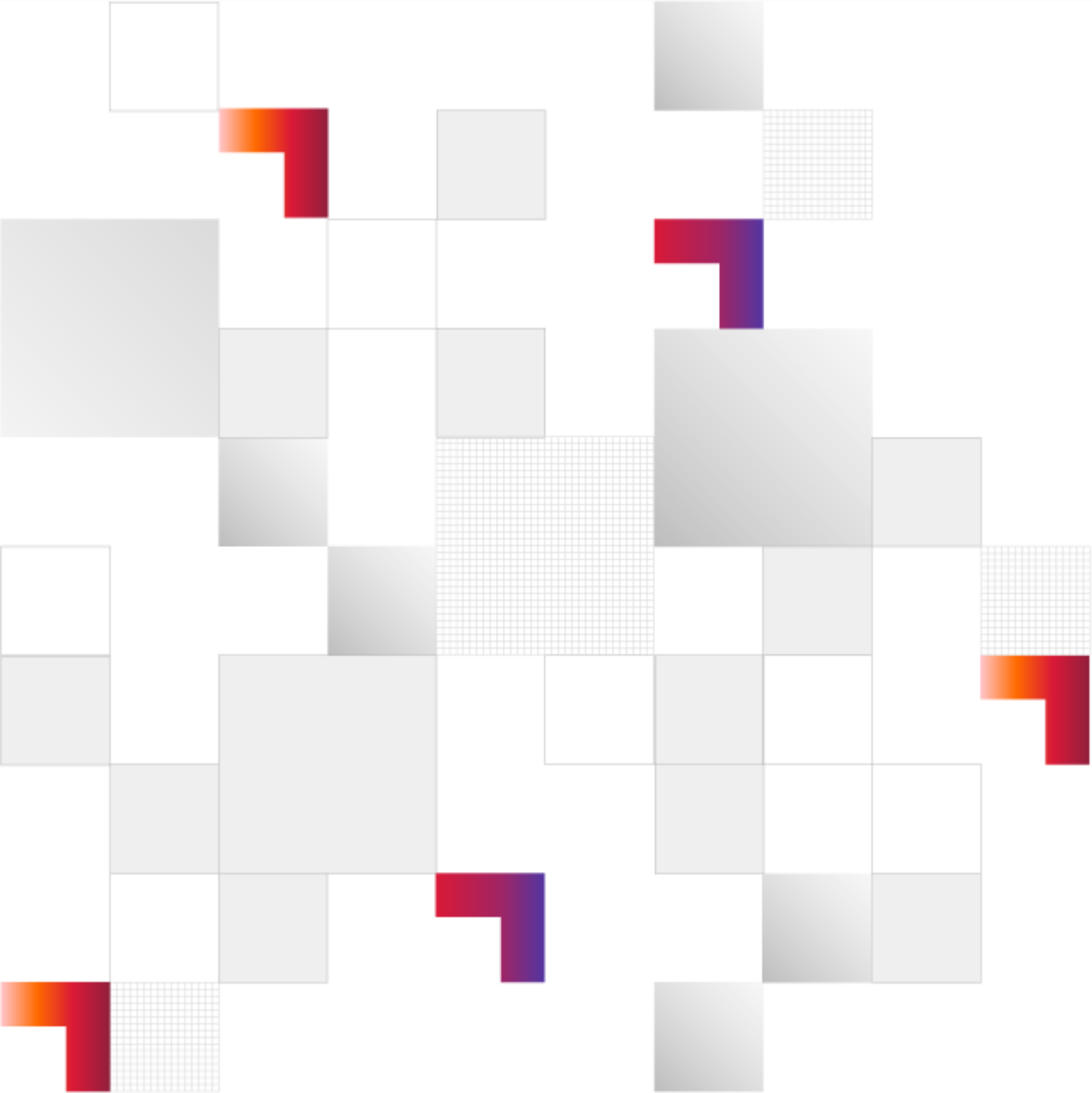


Making Results Accessible

- nobody wanted to read one large Markdown report
 - ⇒ multiple focused report templates
- Markdown and text are still difficult to read
 - ⇒ visualization tool
 - ⇒ integration into a Quality Dashboard
- interaction also possible via MCP server



Results

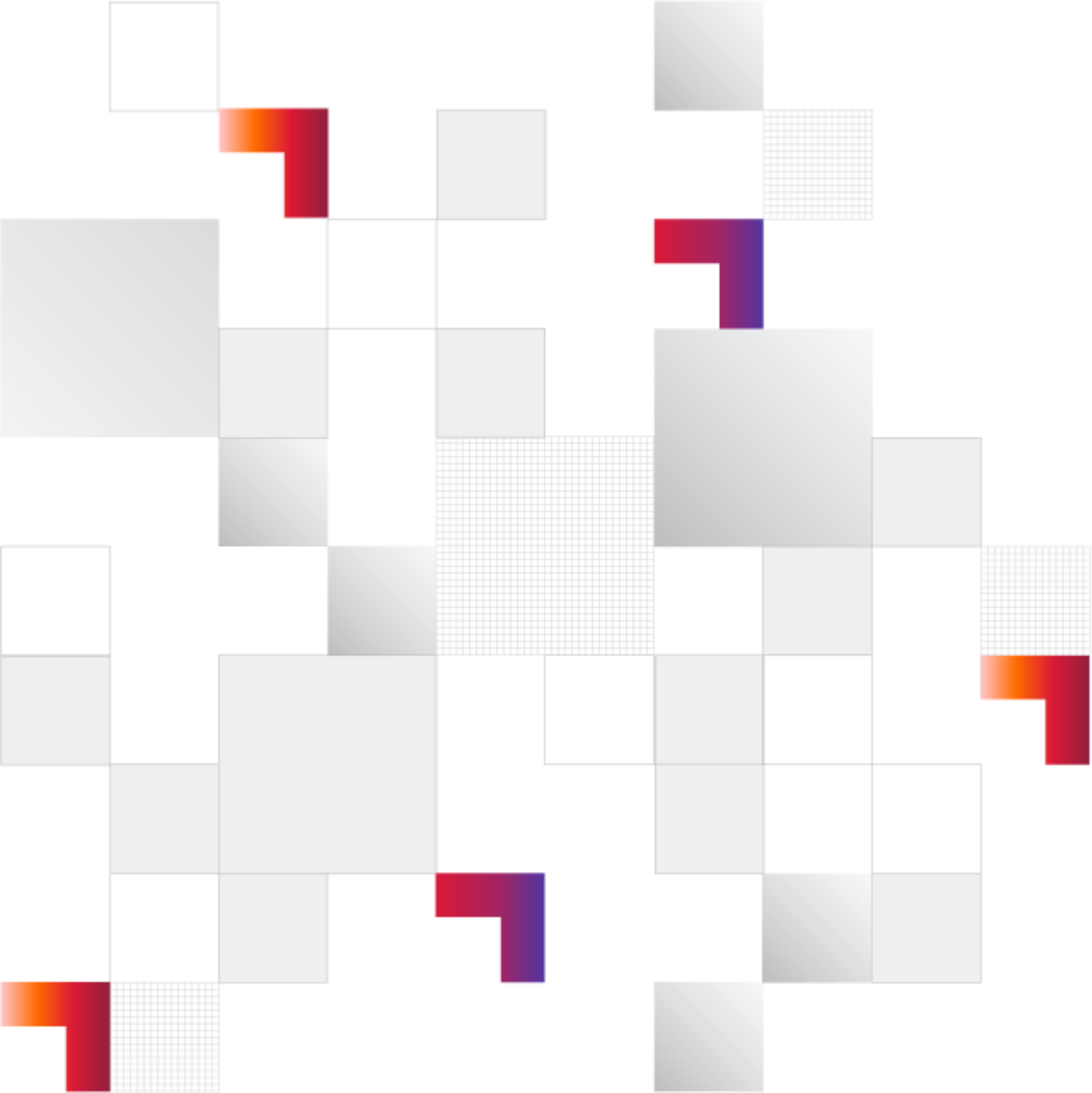


Results

- 2-3 days manual effort
→ approx. 1-2 hours + €20 in tokens
- at 90% review quality
 - developers simply ignore AI errors
- efficient interaction with the results
- almost exclusively positive feedback
- immense value for now 400+ projects



Learnings



What Have We Learned?

- even good prompts are no guarantee
- LLMs tend to follow instructions ... while missing the point
- LLM outputs should be continuously evaluated
- a harness is essential
- No LLM if a script can do it.

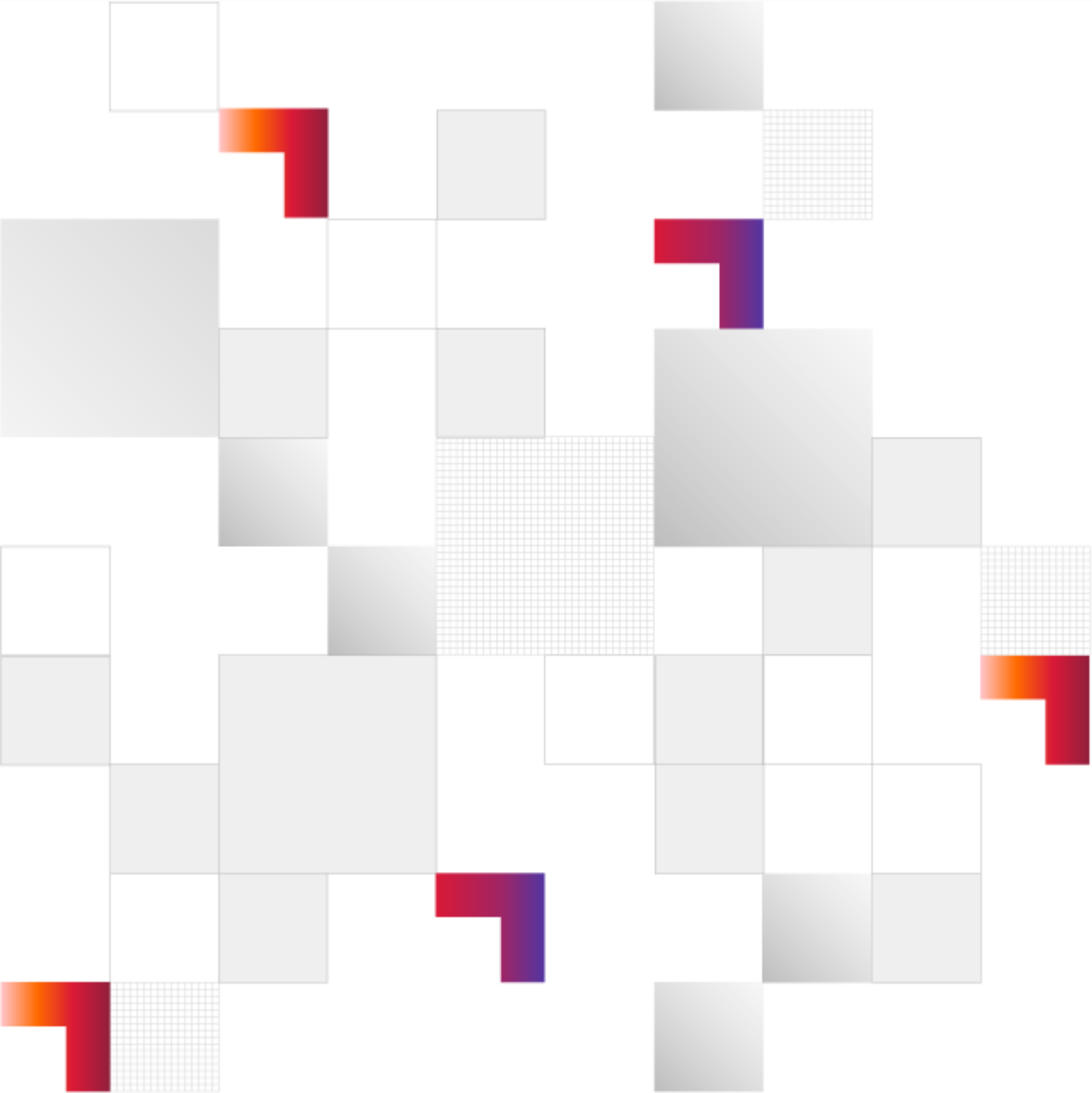


Model Performance

- with Anthropic
 - Haiku → unusable
 - Sonnet → cost/benefit sweet spot
 - Opus → better than Sonnet, but slower and 10x more expensive
- with local models
 - works in principle, but very slow
 - big quality bottleneck: context window
 - (27b models) quality significantly below Sonnet



Next Steps



Next Steps

- evaluation of new models
- tools for deterministic structural analysis
- checklist management
- findings management
- incremental reviews
- create PRs / fix issues automatically

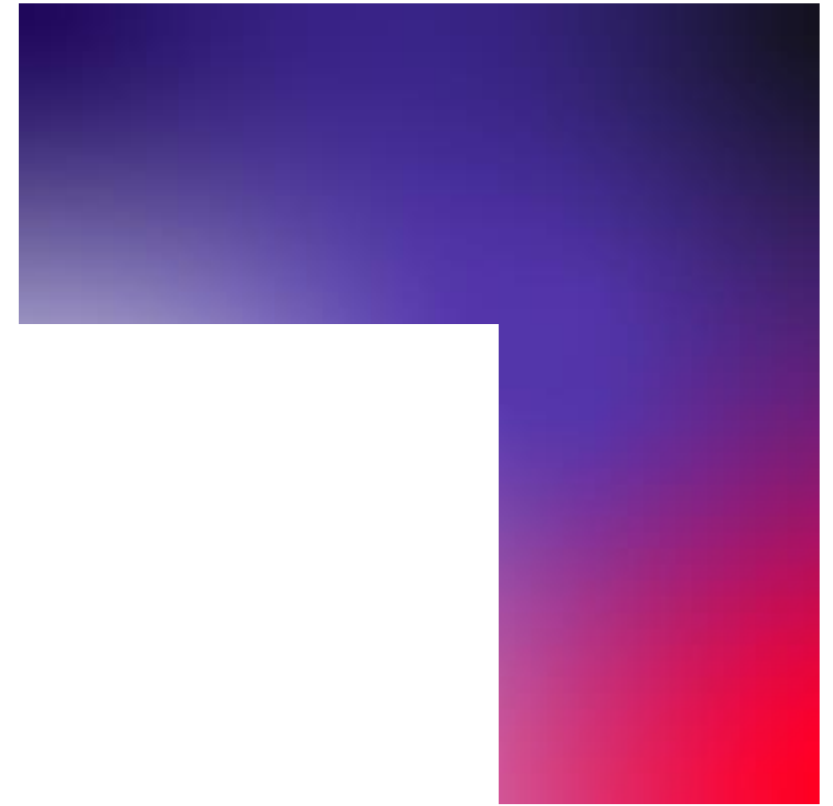


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